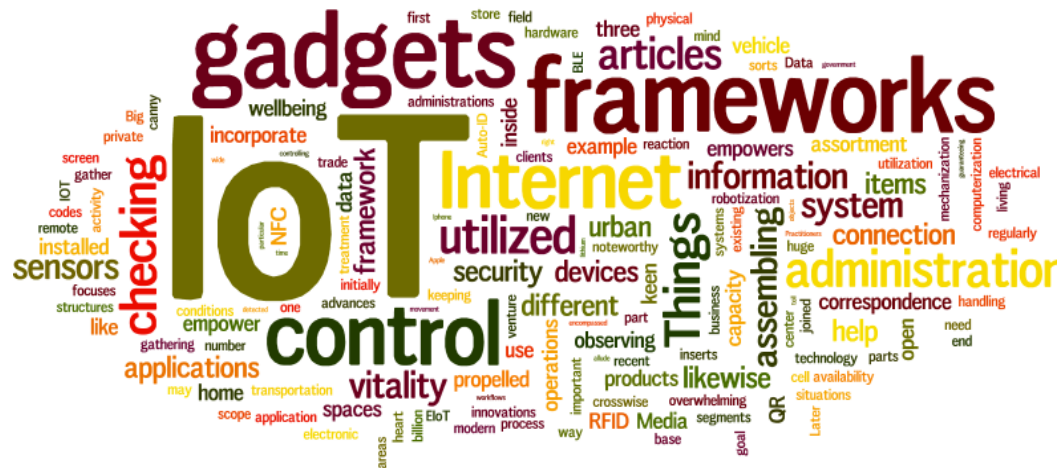


CS578: Internet of Things

Introduction to IoT

Definitions, Characteristics, Applications



Dr. Manas Khatua

Assistant Professor, Dept. of CSE, IIT Guwahati

E-mail: manaskhatua@iitg.ac.in

"I have no special talent. I am only passionately curious." – **Albert Einstein**

What is IoT?

- Internet of Things (IoT) is the network of physical objects – devices, vehicles, buildings and other items – embedded with electronics, software, sensors, and network connectivity that enables these objects to collect and exchange data.
- The basic premise and goal of IoT is to “connect the unconnected”
 - “Things” or objects that were not supposed to be connected to the Internet
- IoT is a technology transition in computer network
 - allow us to sense and control the physical world by making objects smarter and connecting them through network
 - Automation system and connected devices work together to provide a world where data can be collected and analyzed



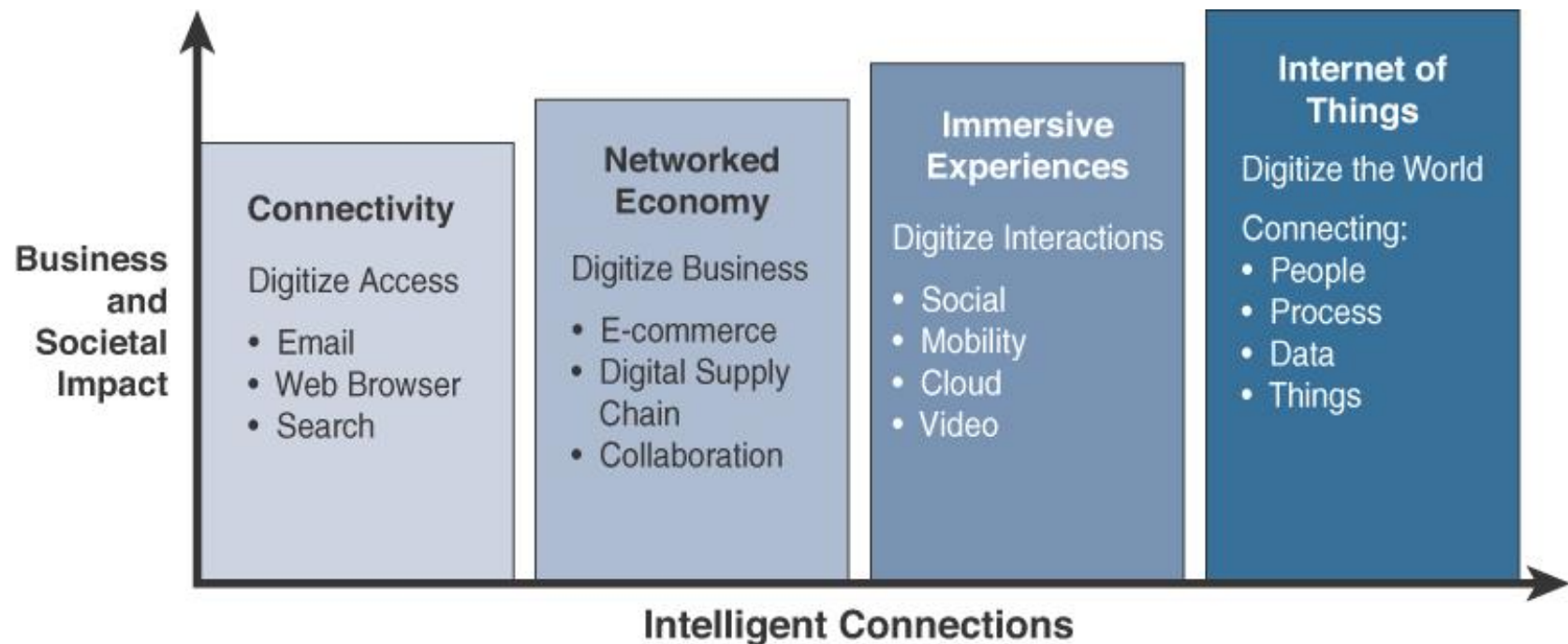
- Unifications of technologies:
 - Embedded systems,
 - Low power and low rate network,
 - Internet,
 - Big data,
 - Data analytics,
 - Cloud computing,
 - Software defined networks,
 - Etc.
- Alternate Definition:

“The Internet of Things (IoT) is the **network of physical objects** that contain embedded technology to **communicate** and **sense** or **interact** with their internal states or the external environment.” – Gartner Research *

* <https://www.gartner.com/en/information-technology/glossary/internet-of-things>

Genesis of IoT

- The term "**Internet of things**" was likely coined by **Kevin Ashton** of Procter & Gamble, later MIT's Auto-ID Center, in 1999.
- “In the twentieth century, **computers were brains without senses** — they only knew what we told them.” IoT is changing this paradigm; in the twenty-first century, **computers are sensing things for themselves!** – Kevin Ashton



Evolutionary Phases of the Internet

Benefits of IoT

❑ To the End User

- **Ease of Doing Things**

- **Emergency Services**

- **Security Services**

❑ To the Businesses

- **Process Improvement**

- **Asset Utilization**

- **New Market Opportunities**

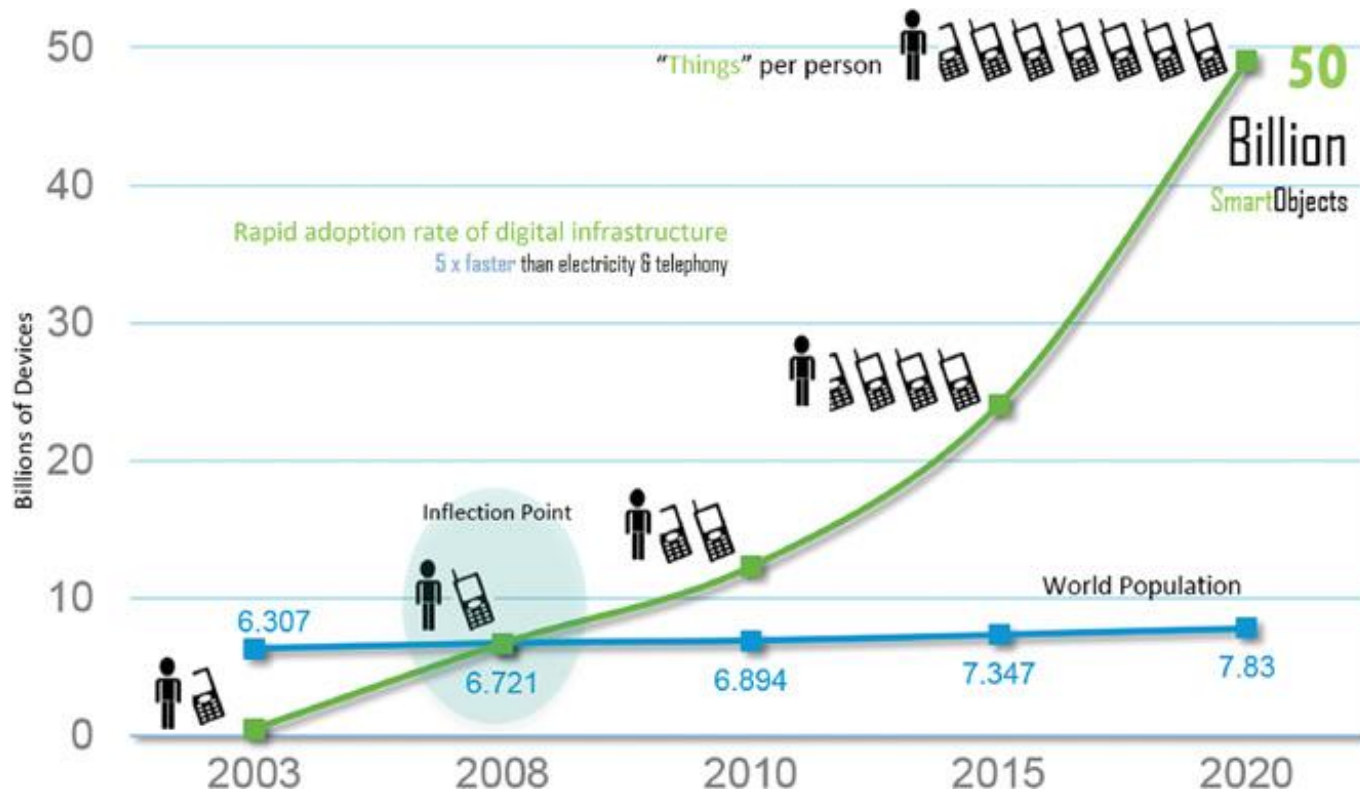
- **Workplace Security**

IoT vs. WSN



- **Wireless Sensor Network (WSN):**
 - WSN refers to a group of specialized dedicated **sensors** with a communications infrastructure.
 - WSN is primarily used for **monitoring** and **recording** the physical environment conditions like temperature, sound, pollution levels, humidity, wind, and so on.
 - It is designed to acquire, process, transfer, and provide data/information extracted from the physical world.
 - In a WSN, there is **no direct connection** to the internet. Instead, the various sensors connect to some kind of router or central node.
- **WSN:** Resource constraint sensor nodes + wireless network to connect the nodes + gather some data by sensing the environment.
- **IoT:** WSN + Internet + App + Cloud computing + Data Analytics + etc...

Growth of IoT Devices vs. Global Population



Cisco System's prediction

- In 2017, Cisco Systems estimates that these new connections will lead to **\$19 trillion** in profits and cost savings

Where is IoT?

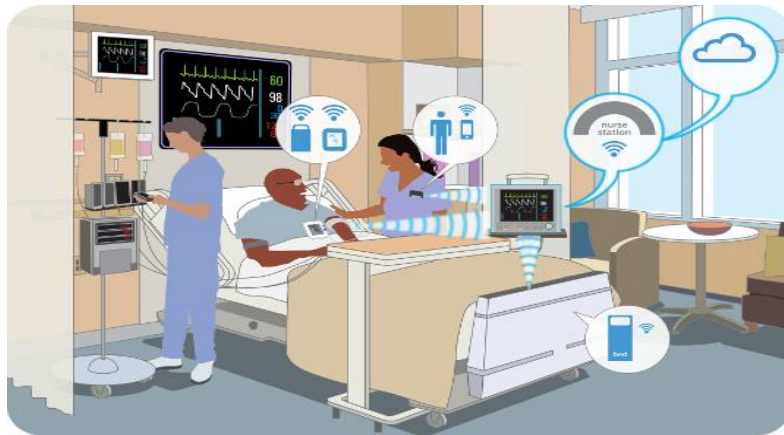


Wearable
Tech Devices



Smart Appliances

It's everywhere!

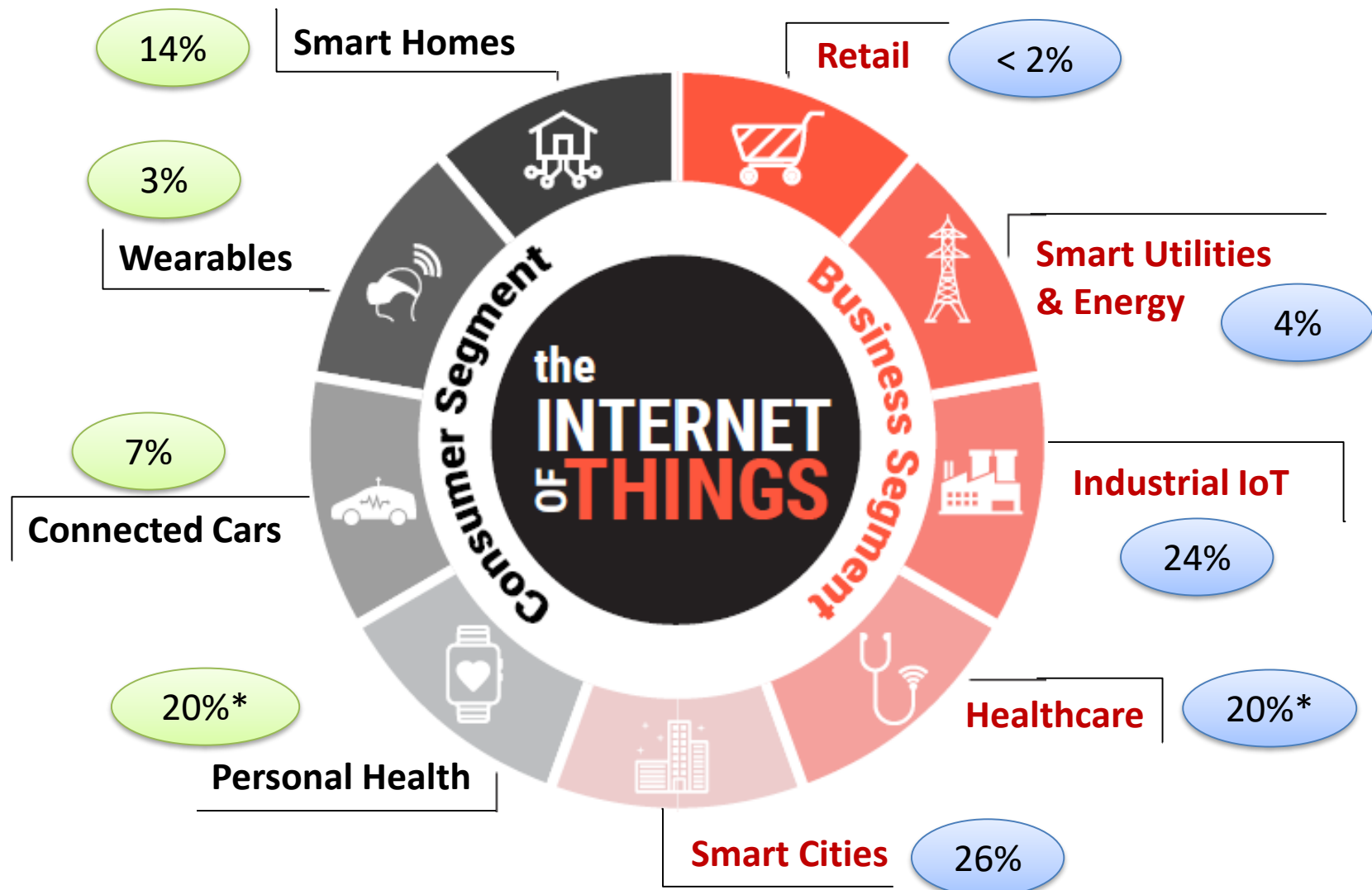


Healthcare



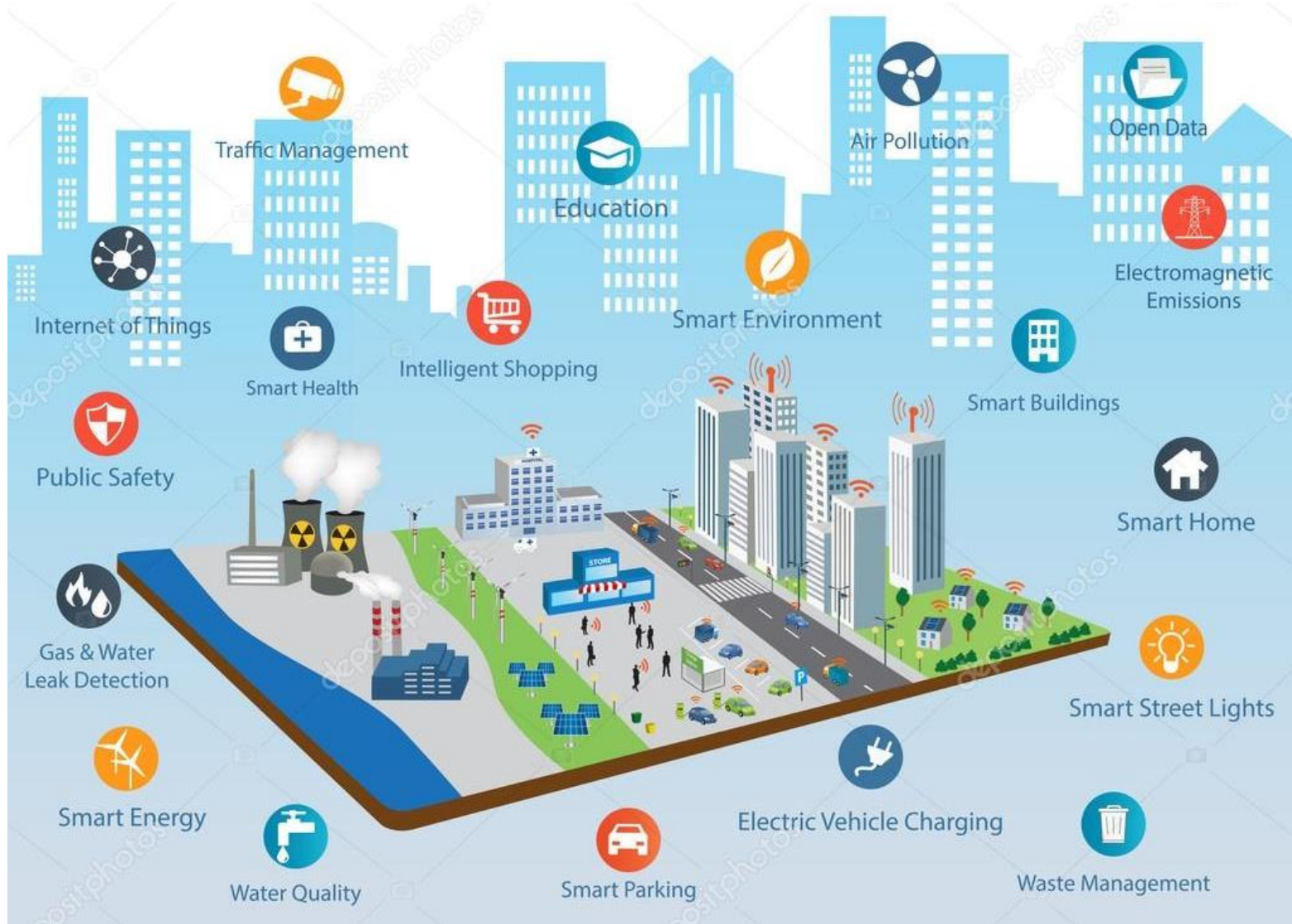
Industry Automation
and Monitoring

Global IoT Market Share



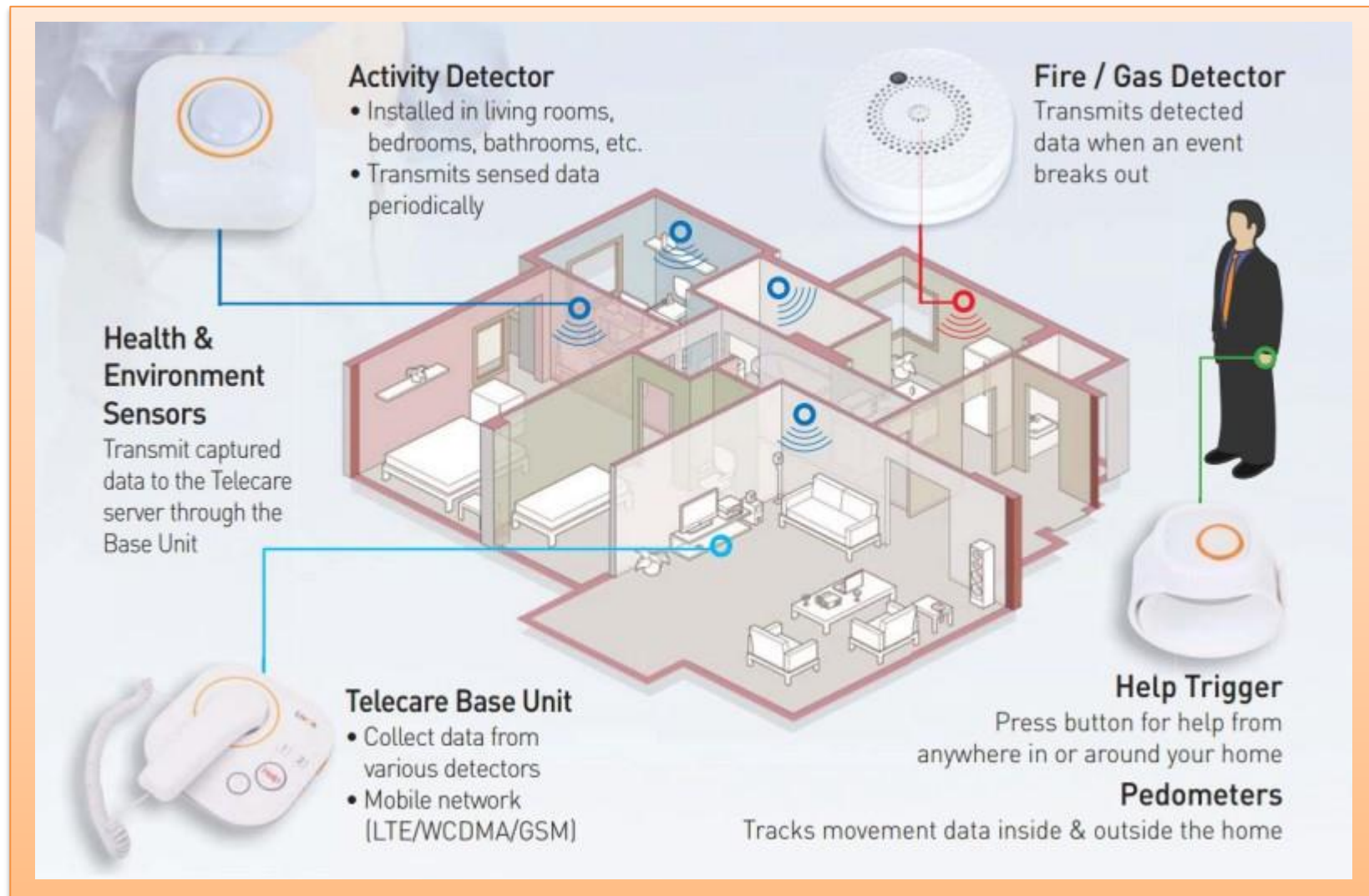
Source: <https://growthenabler.com/flipbook/pdf/IOT%20Report.pdf>

Smart City



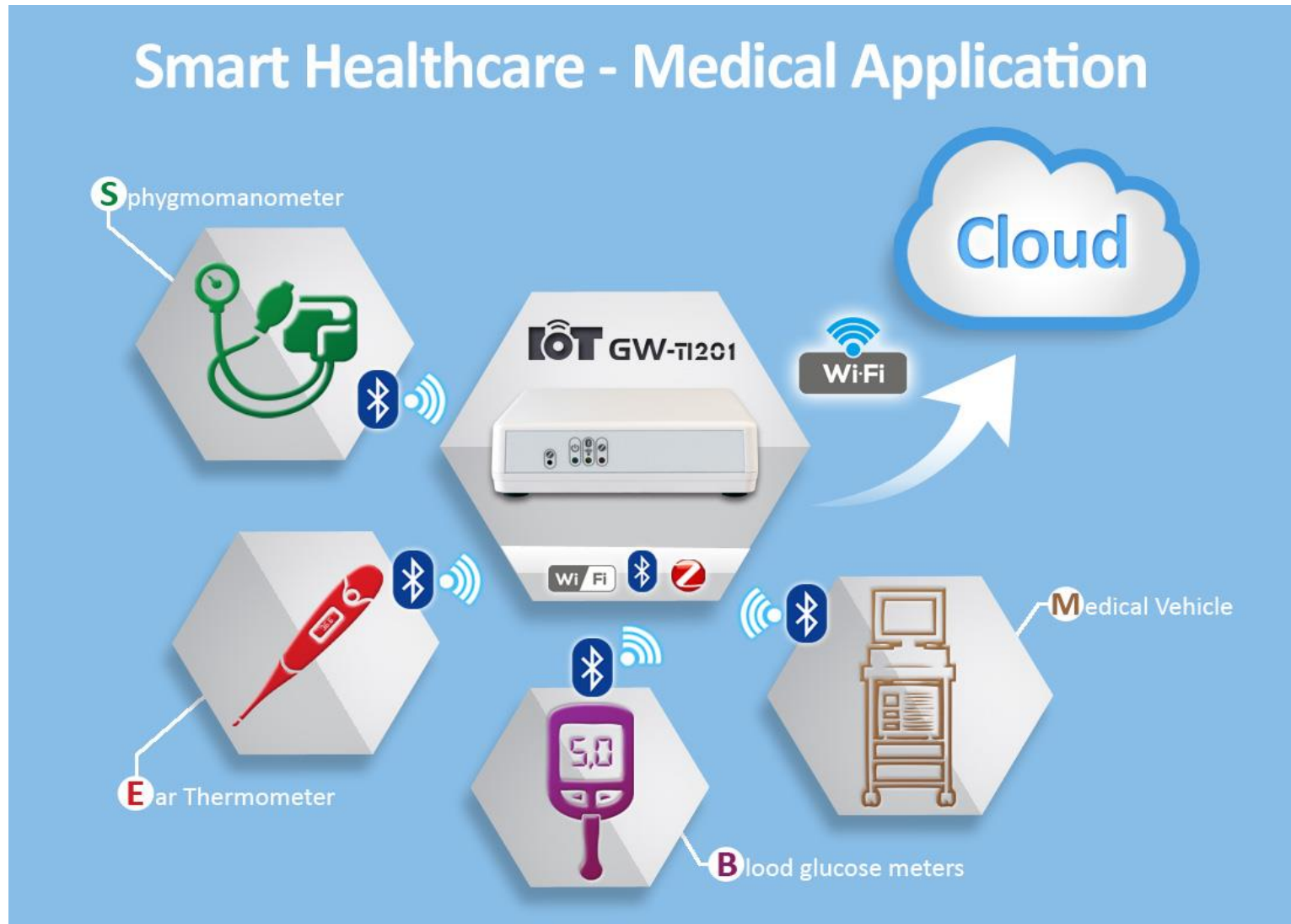
Source: <https://depositphotos.com/126025652/stock-illustration-smart-city-concept-and-internet.html>

Smart Home



Source: <https://medium.com/@globalindnews/north-america-accounted-for-major-share-in-the-global-smart-home-healthcare-market-in-2015-cc9cc1974ac5>

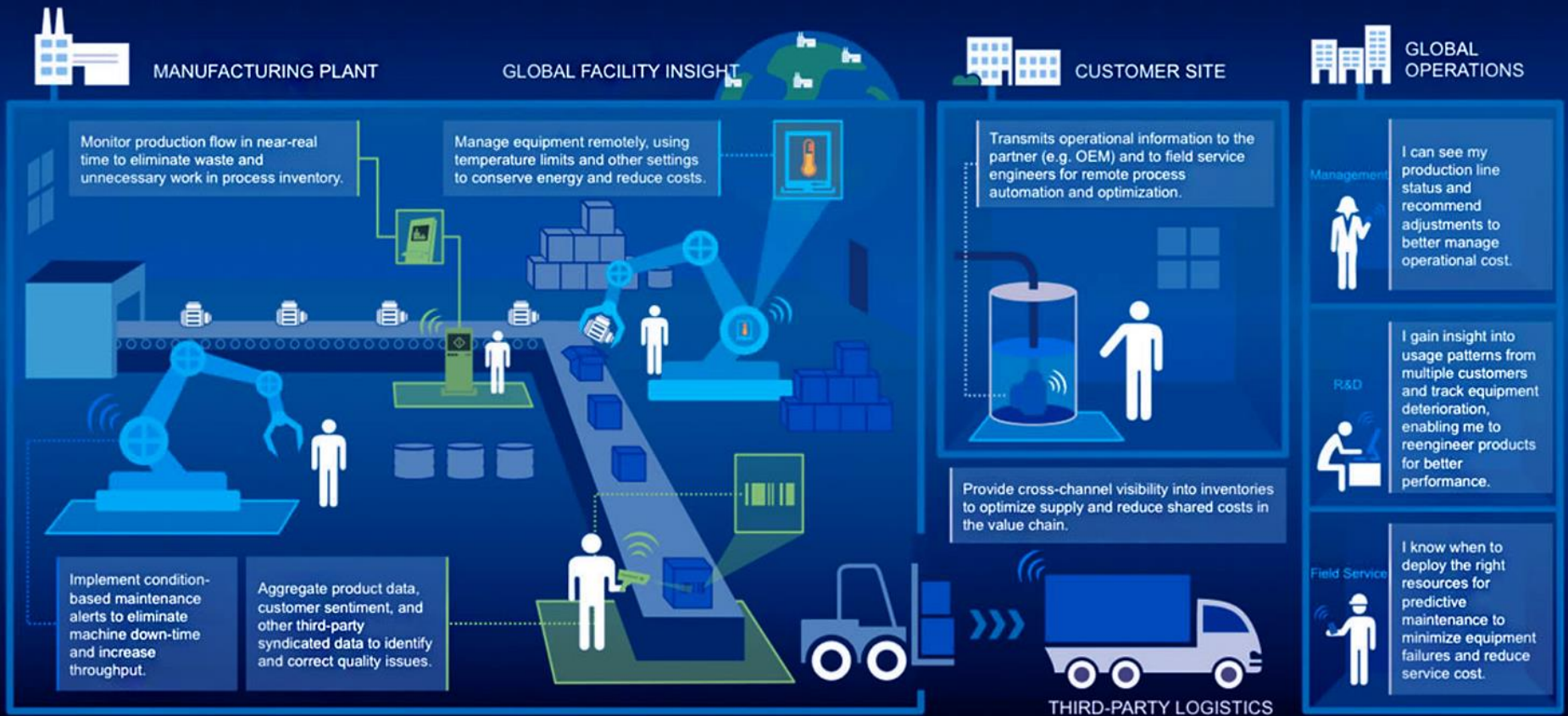
Smart Healthcare



Source: <http://iot.fit-foxconn.com/>

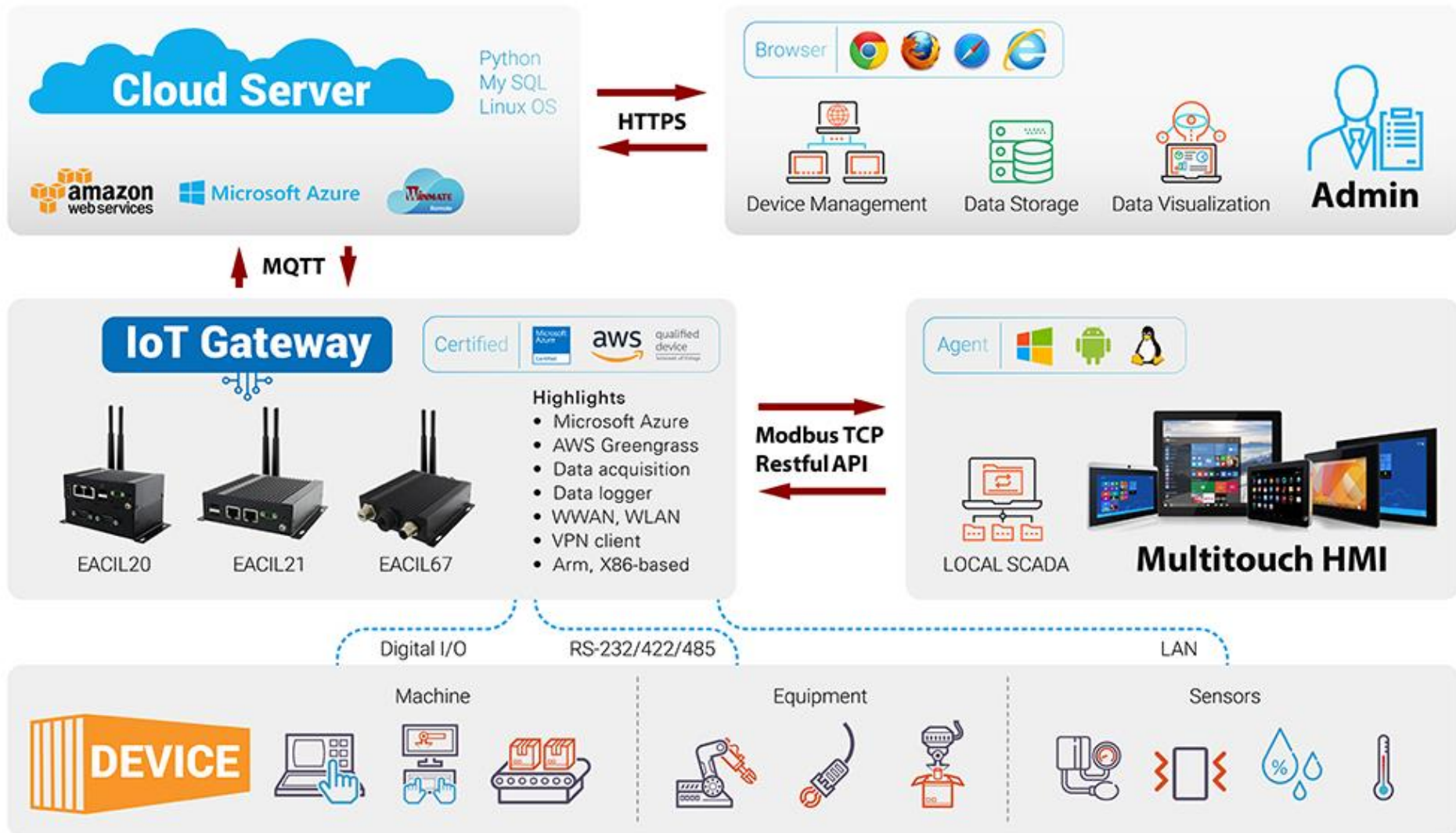
Smart Manufacturing

Internet of Things in Manufacturing



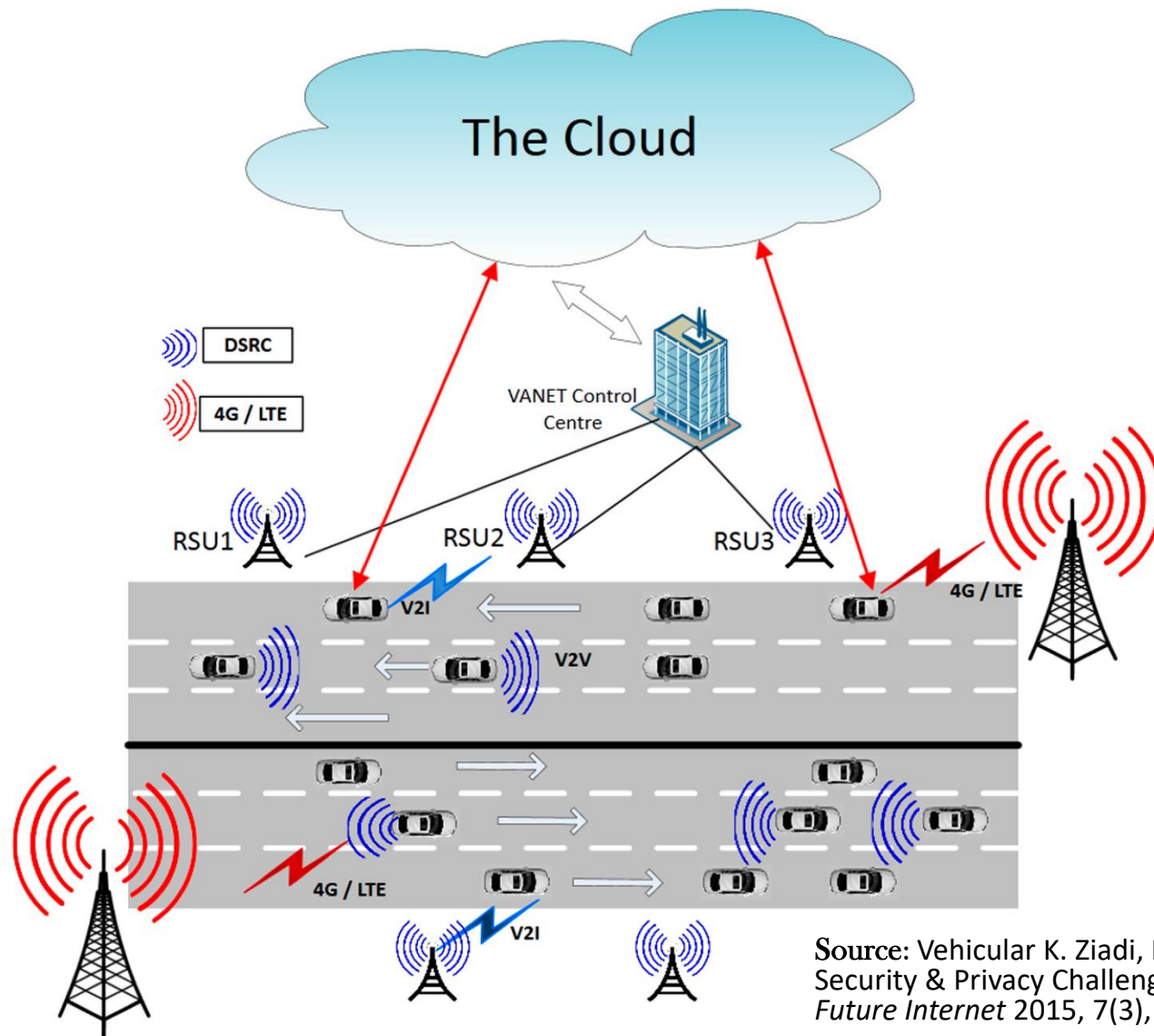
Source: <https://data-flair.training/blogs/industrial-iot-applications/>

Industrial IoT



Source: https://www.winmate.com/Solutions/Solutions_IoT.asp

Connected Cars



Source: Vehicular K. Ziadi, M. Rajarajan, "Internet: Security & Privacy Challenges and Opportunities", *Future Internet* 2015, 7(3), 257-275.

Google's Self-Driving Car



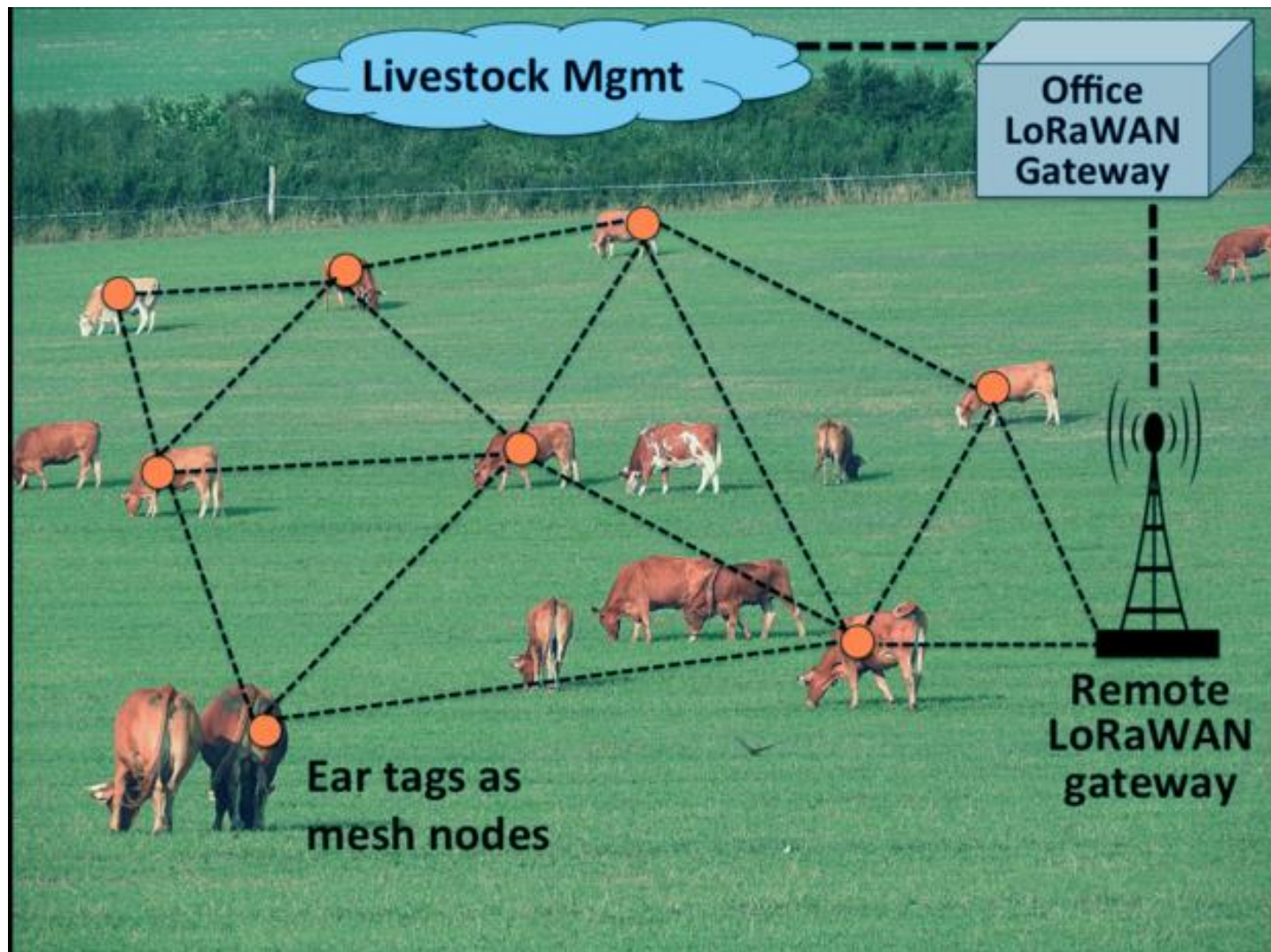
Source: <https://www.google.com/>

Smart Agriculture



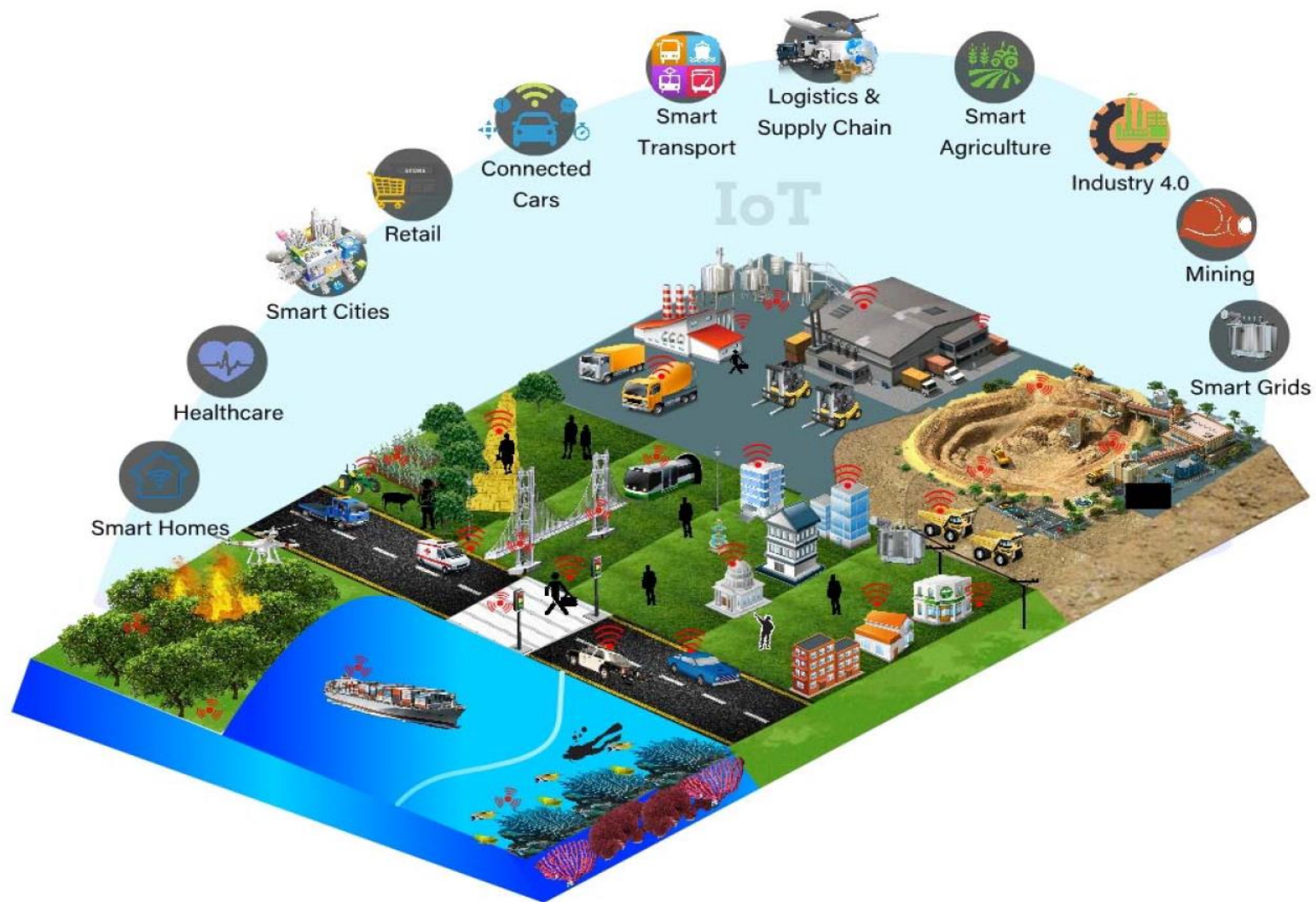
Source: <https://in.pinterest.com/pin/515380751093603767/?lp=true>

Livestock Management



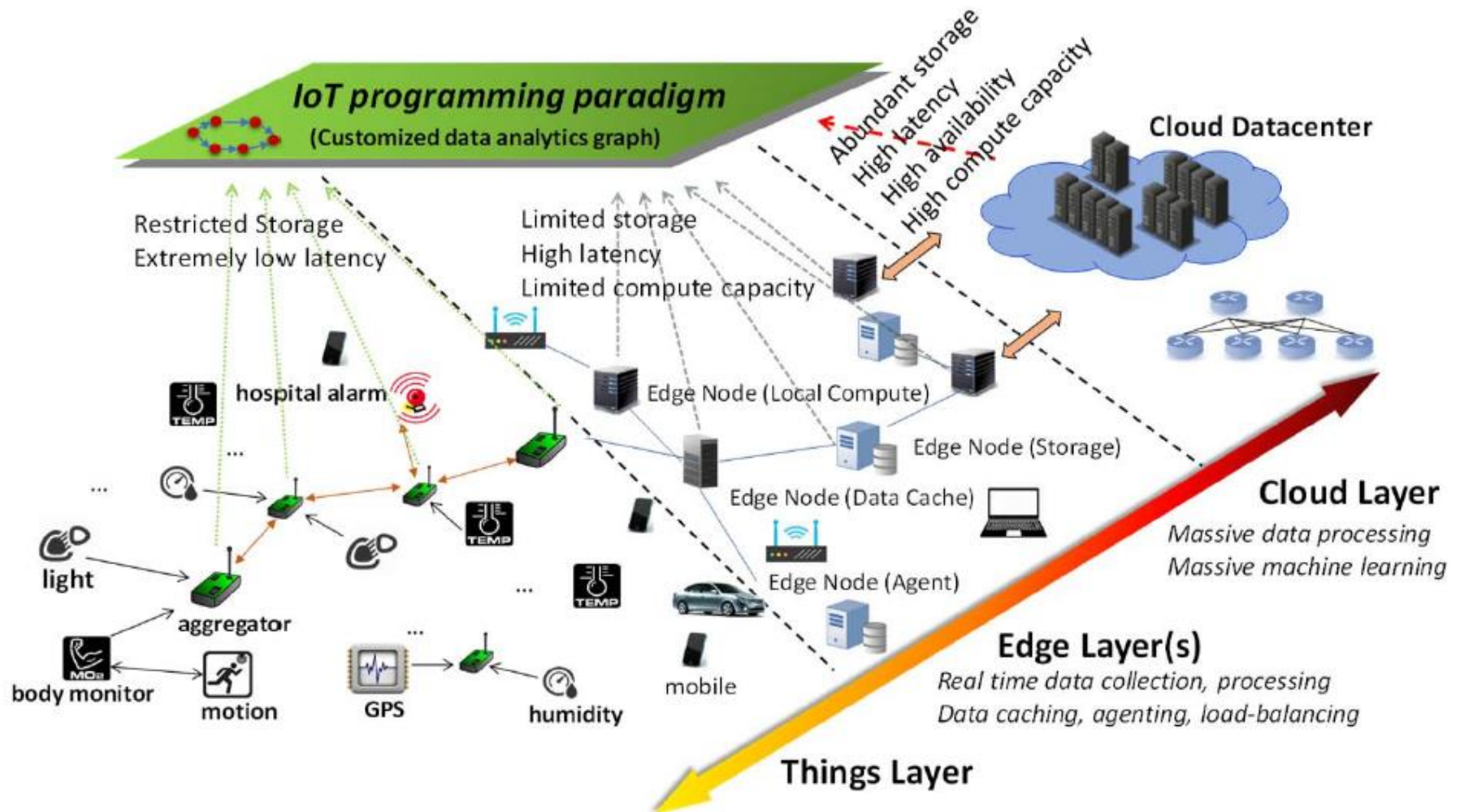
Source: <https://data-flair.training/blogs/iot-applications-in-agriculture/>

Many More



Source: Rajiv Ranjan *et. al.*, "Integrating the IoT and Data Science" *IEEE Cloud Computing*, 2018

How IoT works?



Source: Rajiv Ranjan *et. al.*, "Integrating the IoT and Data Science" *IEEE Cloud Computing*, 2018

Main Challenges in IoT

Sensors

- Limited resources
- Limited types of sensors

Scale

- millions of devices are connected to form IoT

Privacy

- which personal data to share with whom
- how to control

Security

- “things” becomes connected, so security becomes complex

Low Power Network

- Devices should remain connected to the Internet for years
- High network latency

Big data and Data analytics

- massive amount of sensor data
- different sources and various forms
- extract intelligence from the heaps of data

Interoperability

- various protocol, various architecture
- unavailability of standardized platform
- different technology leads to interoperability issue
- Recent IoT standards are minimizing this problem

Lessons Learned



- ✓ Learned about what is IoT
- ✓ Learned the genesis of IoT
- ✓ Understand the benefits of IoT
- ✓ Learned about the market share of IoT
- ✓ Understand the real world applications of IoT
- ✓ Understand various challenges IoT implementation is facing

Thanks!



Figures and slide materials are taken from the following Books:

1. David Hanes *et al.*, “**IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things**”, 1st Edition, 2018, Pearson India.
2. Mayur Ramgir, “**Internet of Things: Architecture, Implementation and Security**”, 1st Edition, 2020, Pearson India.